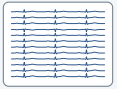
Study pathway

EchoNext paired cohort



12-lead ECG



Tabular inputs

Split	Records	Patients
Train	72,475	26,218
Validation	4,626	4,626
Test	5,442	5,442

Predefined patient-level split

No patient overlap

Masked pretraining

Shared temporal patches across leads

Encoder transfer

Downstream SHD detection

ECG + Tabular inputs

12 prediction targets

11 abnormalities + Overall SHD

Fixed test set: 5,442 patients

Evaluation framework

Matched initialization comparison

Full vs Strict

Same architecture, inputs, split and training

Label efficiency

1% 5% 10% 30% 100%

Nested labeled training subsets

Supporting analyses

Modality · targets · reduced leads · subgroups